Bayesian weight estimation for adaptive power control in electromagnetic robotic grippers

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***Abstract*—Electromagnetic robotic grippers can generate large holding forces with compact and direct actuation. However, their grasping force depends strongly on the supplied current, and excessive current increases power consumption and thermal load. This paper presents a Bayesian weight-estimation method for regulating the power of a liquid-cooled electromagnetic robotic gripper during vertical lifting. The proposed method uses load-cell measurements to estimate the object weight while the load is progressively transferred to the gripper. The raw weight measurement is first corrected using an experimentally defined piecewise calibration function. Then, a conservative load prediction is computed from sequential corrected measurements and incorporated into a Bayesian update, producing a posterior weight estimate. Finally, the estimated weight is converted into a normalized LCLE power command through a probabilistic weight-to-power mapping. The method is intended for energy-aware electromagnetic grasping, where the gripper increases power according to the estimated load instead of applying maximum power throughout the entire task. The proposed framework is evaluated using a liquid-cooled laminar electromagnet gripper and objects from 5 kg to 9 kg, showing the feasibility of adaptive weight estimation during vertical lifting and load-dependent LCLE power assignment without requiring the object weight to be known before grasping.**

***Index Terms*—Electromagnetic gripper, Bayesian estimation, force sensing, power regulation, liquid-cooled actuator, heavy object grasping.**

I. Introduction

Robotic grasping of heavy objects remains challenging because the gripper must generate sufficient holding force while maintaining a compact, lightweight, and reliable structure [1], [2]. Conventional rigid grippers and anthropomorphic hands can provide well-defined contact kinematics, but their payload capability is often limited by actuator size, transmission losses, and the difficulty of integrating high-force mechanisms into compact fingers [7]–[9]. Electromagnetic grippers offer an alternative because they can generate direct attractive forces on ferromagnetic targets without requiring gears, tendons, screws, or other mechanical transmissions at the contact interface [11], [15], [16]. However, rigid electromagnetic grippers are usually limited to objects with suitable metallic contact surfaces and tend to provide little passive adaptation to geometric variations [11]. In contrast, soft or compliant electromagnetic grippers can improve adaptability by allowing the fingers or contact structures to conform to different object poses and surface geometries [1], [4]. This compliance is beneficial for grasping,

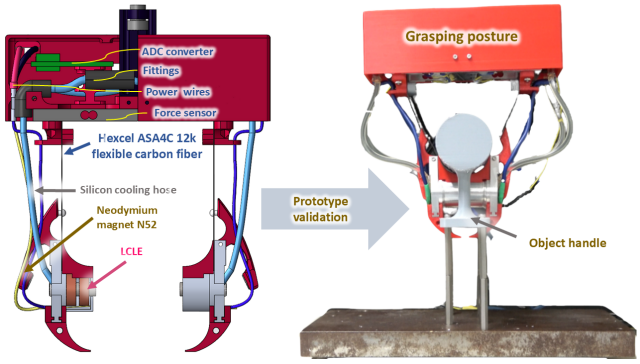


Fig. 1. Liquid-cooled electromagnetic gripper used as the experimental platform for Bayesian weight estimation and power regulation. The LCLE fingertips generate the grasping force, while the integrated force-sensing structure measures the load transmitted by the object during vertical lifting.

but it also makes control more difficult because the final grasping posture, contact area, and load transfer can vary from trial to trial [1], [3].

The liquid-cooled electromagnetic gripper used in this work is shown in Fig. 1. The gripper was originally developed as a compliant platform for heavy-object grasping using Liquid-Cooled Laminar Electromagnets (LCLEs) as active fingertips [1], [2]. In the present paper, the gripper structure and the LCLE actuator are not redesigned. Instead, the same platform is used to study Bayesian weight estimation and power regulation during vertical lifting.

Previous electromagnetic grippers and robotic fingers have mainly used electromagnets for joint locking, braking, self-adaptive grasping, repeatability improvement, or low-power holding/release mechanisms [4], [6]–[9]. Magnetic actuation has also been applied to micromanipulation, magnetic levitated fingers, variable-stiffness devices, and compliant finger mechanisms [10], [15], [16]. However, these studies mainly address precise motion generation, stiffness control, mechanical design, or mode switching, rather than actuation-power regulation for heavy-object grasping [4], [6]–[10].

This limitation becomes more relevant in heavy-object manipulation. Previous electromagnetic grippers and robotic fingers often use magnets for contact, locking, braking, or sensing, while the main grasp is maintained mechanically or by another actuator [4], [7]–[9], [11]. However, these

approaches do not primarily address progressive power regulation for soft electromagnetic grippers during heavy-object lifting.

Our previous LCLE-based gripper demonstrated high holding forces for heavy-object grasping [1], [2]. However, its conservative high-initial-power strategy increased electrical demand and thermal load [1], motivating the development of a load-dependent power regulation method that progressively adjusts the LCLE input during lifting.

In soft electromagnetic grippers, load measurements vary during lifting due to compliance, contact deformation, and load transfer [1], [3]. Unlike fixed-power operation, weight-dependent control adjusts the LCLE input progressively according to the estimated load.

This paper proposes a Bayesian weight-estimation and power-regulation method for a liquid-cooled electromagnetic robotic gripper, see Fig. 1. The method uses load-cell measurements to obtain a corrected weight value, predicts the future load during lifting, updates the object-weight estimate through a Bayesian formulation, and maps the estimated weight into a normalized power command for the LCLEs. Unlike previous electromagnetic grippers that mainly emphasize mechanical design, contact repeatability, braking, sensing, stiffness variation, or mode switching, the proposed method focuses on how the electromagnetic power should be assigned during heavy-object lifting [4], [6]–[9], [11]. The objective is not to redesign the LCLE actuator or the gripper structure, but to improve the way the existing liquid-cooled electromagnetic gripper is operated when the object weight is initially unknown.

II. Electromagnetic Gripper and Force-Sensing System

A. Experimental platform

The experimental platform is a two-finger liquid-cooled electromagnetic gripper derived from our previous LCLE-based heavy-object grasping studies [1], [2]. Unlike those works, this paper does not address gripper or actuator design; instead, the existing LCLE-based gripper is used to evaluate Bayesian weight estimation and power regulation during vertical lifting.

The electromagnets are cooled using dielectric oil circulated through a closed-loop cooling system. This allows the actuator to operate under high-current conditions while reducing the winding temperature increase. Since the objective of this paper is power regulation, the cooling system and the LCLE actuator are treated as the hardware basis that enables repeated lifting experiments under different power levels.

B. Load-cell-based weight measurement

The gripper integrates a force-sensing structure based on load-cell measurements. Two sensing signals are acquired using analog-to-digital conversion and averaged to estimate the load transmitted to the gripper body. The force-sensing structure is shown in Fig. 2. Strain gauges are attached to deformable beams in the gripper body, allowing the transmitted

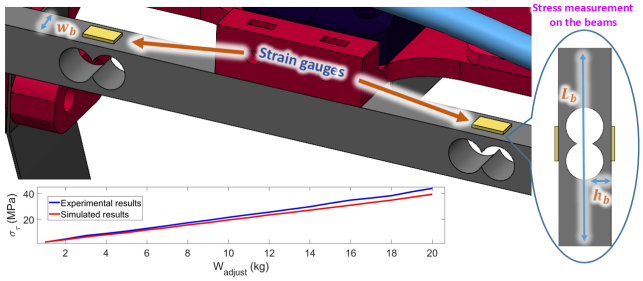


Fig. 2. Force-sensing structure used for weight estimation. Strain-gauge measurements are compared with simulated stress responses as a function of the adjusted weight W_{adjust}, corresponding to the corrected weight W_c.

load to be estimated from the measured deformation. The sensor response was evaluated experimentally and compared with simulated results over the calibrated weight range. Let w_D and w_I be the right and left load-cell readings, respectively. The mean sensor signal is calculated as

$$M_e = \frac{w_D + w_I}{2}. \tag{1}$$

The measured signal is converted into an initial weight estimate through an empirical calibration function,

$$W_m = f_s(M_e), \tag{2}$$

where W_m is the measured weight and $f_s(\cdot)$ represents the conversion from the load-cell signal to kilograms. In practice, this conversion includes offset compensation and experimentally determined scaling factors. The purpose of this first calibration stage is to extend the useful measurement range of the sensing system and provide a weight value that can be used by the controller. In the sensor characterization results shown in Fig. 2, the corrected weight is denoted as W_{adjust}; in the remainder of this paper, the same corrected quantity is written as W_c for compactness and consistency with the Bayesian estimation formulation. Although this first calibration provides a usable weight estimate, the measured weight may still contain load-dependent bias. This effect becomes relevant because the gripper includes flexible elements, and its real operating behavior depends on the object weight, contact condition, grasping posture, and supplied power.

Therefore, the initial value is further corrected using a piecewise calibration function. The corrected weight is written as

$$W_c = W_m + \Delta W(W_m), \tag{3}$$

where $\Delta W(W_m)$ is a lookup-table correction selected according to the measured weight range.

Let $\mathbf{T}_l$ be the vector of weight thresholds and $\mathbf{T}_c$ the vector of correction values obtained from calibration experiments. For a positive measured weight, the correction index is defined as

$$idx = \min\{i : \mathbf{T}_l(i) > W_m\}. \tag{4}$$

Then, the corrected weight is calculated as

$$W_c = \begin{cases} 0, & W_m < 0, \\ W_m + \mathbf{T}_c(idx-1), & W_m \geq 0. \end{cases} \tag{5}$$

The correction index is defined for nonnegative measurements within the calibrated range, with the first threshold set to zero. Measurements outside this range require separate boundary handling. The empirical correction was obtained from controlled calibration experiments because the gripper response is difficult to model analytically under flexible deformation, contact variations, and load-dependent effects. under flexible deformation, contact variations, and load-dependent effects. During calibration, the object, grasping configuration, power-supply setup, and manipulation conditions were kept constant, while only the object weight was progressively increased to characterize the supported-weight/input-power relationship. The corrected weight W_c is then used as the input measurement for the Bayesian estimator, allowing the controller to rely on a calibrated value that better represents the gripper behavior under the tested manipulation conditions instead of using the raw load-cell measurement directly.

C. Weight-to-power calibration

After obtaining the corrected weight W_c, the controller maps the estimated load to a normalized LCLE power command. Each calibration weight was tested at least five times to verify the consistency of the minimum power required to maintain the object suspended. In the experimental range of 5–9 kg, the required normalized power remains relatively low, approximately between 7.5% and 11.7%, avoiding maximum-power operation from the beginning of the grasp. During calibration, the object geometry, grasping posture, power-supply configuration, and lifting procedure were kept constant while the object weight was progressively increased. This characterized the gripper operating region without requiring a purely analytical model of the flexible grasping interaction.

D. Control procedure

During each lifting trial, the controller acquires the two load-cell signals, computes the mean value M_e, and converts it into an initial weight measurement W_m. A piecewise correction is then applied to obtain W_c, which is used as the input to the Bayesian estimator. At each sampling instant, the current and previous corrected measurements are used to compute the conservative prediction W_k^-, which is combined with the prior distribution to obtain the posterior estimate $\hat{W}_k$. This estimate is then mapped into a normalized LCLE power command, allowing the gripper to progressively increase the input power according to the estimated object weight instead of applying maximum power at the beginning of the grasp.

III. Bayesian Weight Estimation

A. Measurement Model

The corrected sensor measurement at the k-th sampling instant is denoted by y_k. It is modeled as

$$y_k = W + \eta_k, \tag{6}$$

where W is the true object weight and η_k represents the measurement uncertainty. This uncertainty includes sensor noise, calibration error, contact changes, and transient effects during lifting.

During the lifting phase, the object load is progressively transferred from the support surface to the gripper. Therefore, the measured weight may increase over time until the object is fully suspended. To avoid underestimating the load during this transition, a conservative prediction is computed before applying the Bayesian update.

B. Weight Estimation

Let y_{k-1} and y_k be two consecutive corrected weight measurements, and let

$$\Delta t_k = t_k - t_{k-1} \tag{7}$$

be the sampling interval. The local rate of change of the corrected weight is approximated as

$$\dot{y}_k = \frac{y_k - y_{k-1}}{\Delta t_k}. \tag{8}$$

A conservative future-weight prediction is then computed as

$$W_k^- = y_k + \dot{y}_k \Delta t_p, \tag{9}$$

where Δt_p is a short prediction horizon. In the implementation, Δt_p can be selected equal to the sampling interval, i.e., $\Delta t_p = \Delta t_k$, which gives a one-step-ahead prediction. This formulation explicitly represents the prediction horizon instead of implicitly using the previous sampling interval.

To avoid underestimating the load during lifting, the predicted value is constrained to be no smaller than the current corrected measurement:

$$W_k^- = \max(W_k^-, y_k). \tag{10}$$

Negative predictions are also rejected by applying

$$W_k^- = \max(W_k^-, 0). \tag{11}$$

The conservative prediction extrapolates the corrected weight using consecutive measurements while preventing the predicted value from falling below the current measurement. However, this constraint does not guarantee a monotonically increasing estimate, and transient sensor fluctuations may produce oscillations in the predicted weight, as observed in Fig. 5. The Bayesian update subsequently combines this prediction with the prior weight estimate.

C. Bayesian Update

The object weight is treated as a random variable. The prior distribution at the k-th step is assumed Gaussian:

$$W \sim \mathcal{N}(\mu_k^-, (\sigma_k^-)^2), \tag{12}$$

where μ_k^- is the prior mean and σ_k^- is the prior standard deviation.

The conservative prediction W_k^- is used as the observation for the Bayesian update. The likelihood is modeled as

$$W_k^- \mid W \sim \mathcal{N}(W, \sigma_l^2), \tag{13}$$

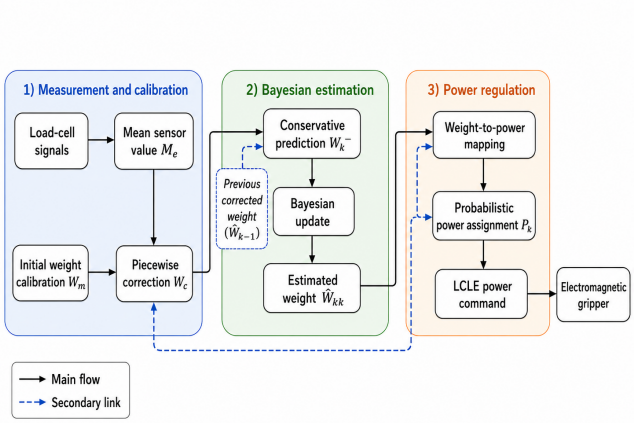


Fig. 3. Proposed Bayesian power-regulation framework. Corrected load-cell measurements are processed through a Bayesian update and mapped into a normalized LCLE power command.

where σ_l is the standard deviation associated with the likelihood. This parameter represents the uncertainty of the corrected and predicted weight measurement.

For a Gaussian prior and Gaussian likelihood, the posterior distribution is also Gaussian. The posterior mean and variance are given by

$$\mu_k^+ = \frac{\sigma_l^2 \mu_k^- + (\sigma_k^-)^2 W_k^-}{(\sigma_k^-)^2 + \sigma_l^2}, \tag{14}$$

$$(\sigma_k^+)^2 = \frac{(\sigma_k^-)^2 \sigma_l^2}{(\sigma_k^-)^2 + \sigma_l^2}. \tag{15}$$

The posterior mean is used as the estimated object weight:

$$\hat{W}_k = \mu_k^+. \tag{16}$$

Finally, the posterior distribution is propagated to the next iteration by setting

$$\mu_{k+1}^- = \mu_k^+, \qquad (\sigma_{k+1}^-)^2 = (\sigma_k^+)^2. \tag{17}$$

The sequential update assumes conditionally independent observations with constant likelihood variance. However, consecutive predictions share sensor measurements and may be correlated. Without process noise, the posterior variance decreases monotonically, potentially leading to overconfidence. Therefore, the resulting uncertainty should be interpreted under these modeling assumptions.

IV. Power Regulation

The complete Bayesian power-regulation framework is summarized in Fig. 3. The corrected load-cell measurement is processed through the Bayesian estimator and then mapped into a normalized LCLE power command.

A. Weight-to-power lookup table

The electromagnetic gripper requires a power command that depends on the estimated object weight. As shown in Table I, a lookup table is experimentally defined to relate weight and normalized power. Let $P \in [0, 1]$ be the normalized power command, where $P = 0$ represents zero input power and $P = 1$ represents the maximum available power.

TABLE I
Empirical weight-to-power calibration used for normalized LCLE power assignment.

Supported weight range (kg)	Power source (%)
Approximately 5 kg	7.5
Approximately 6–7 kg	9.5–10.5
Approximately 8–9 kg	10.5–11.7
Approximately 10 kg	12.7

Given the estimated weight $\hat{W}_k$, the nominal power command is obtained by linear interpolation:

$$P_n(\hat{W}_k) = P_i + \frac{\hat{W}_k - W_i}{W_{i+1} - W_i}(P_{i+1} - P_i), \tag{18}$$

where W_i and W_{i+1} are two neighboring calibration weights, and P_i and P_{i+1} are their corresponding power values. The output is saturated to remain within the valid range:

$$P_n(\hat{W}_k) = \min(1, \max(0, P_n(\hat{W}_k))). \tag{19}$$

B. Probabilistic power assignment

Instead of assigning power only from a single estimated weight value, the proposed method propagates the uncertainty of the weight estimate to the power command. A discrete weight range is considered:

$$\mathcal{W} = \{0, 0.1, 0.2, \ldots, W_{\max}\}. \tag{20}$$

A probability distribution centered at the estimated weight is calculated as

$$p_W(W_j) = \frac{\mathcal{N}(W_j; \hat{W}_k, \sigma_p^2)}{\sum_r \mathcal{N}(W_r; \hat{W}_k, \sigma_p^2)}, \tag{21}$$

where σ_p determines the uncertainty window used for power assignment.

The probability associated with each candidate power value is obtained by accumulating the probability of weights that map to similar power levels. The final power command is calculated as the expected power:

$$P_k = \sum_q P_q p_P(P_q), \tag{22}$$

where $p_P(P_q)$ is the probability assigned to the candidate power P_q. This formulation incorporates weight-estimation uncertainty into the power assignment rather than relying exclusively on a single estimated weight.

V. Experimental Evaluation

A. Experimental procedure

The proposed method was evaluated using the LCLE-based gripper during vertical lifting experiments. Five object weights were considered: 5, 6, 7, 8, and 9 kg. Each weight was tested at least three times to verify the repeatability of the proposed estimation and power-regulation procedure. For each case, the same object geometry, grasping posture, lifting

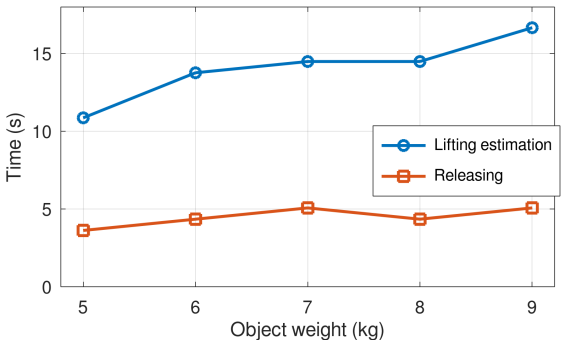


Fig. 4. Effect of object weight on lifting-estimation and releasing times. Lifting time increases with object weight, while the return phase shows a weaker load dependence.

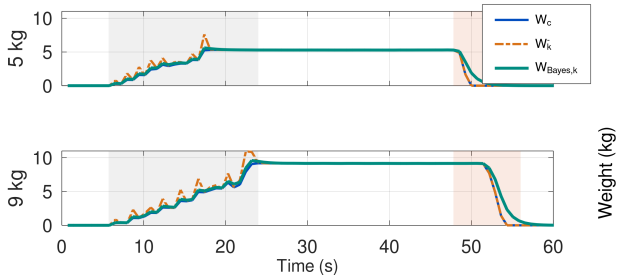


Fig. 5. Representative temporal responses for the 5 kg and 9 kg objects. The corrected weight W_c, conservative prediction W_k^-, and Bayesian estimate $W_{\mathrm{Bayes},k}$ are shown during lifting, holding, and return. In the control formulation, $W_{\mathrm{Bayes},k}$ corresponds to $\hat{W}_k$. The gray region indicates the lifting estimation phase, while the orange region indicates the releasing phase.

configuration, and power-supply setup were used. Only the object weight was modified between trials. This procedure was selected to isolate the effect of the object weight on the Bayesian estimation and power-regulation process. During each experiment, the gripper approached the object handle, closed the electromagnetic fingertips, lifted the object vertically, and then returned it to its original position. During lifting, the force-sensing structure measured the transmitted load, which was processed through calibration, piecewise correction, conservative prediction, Bayesian estimation, and weight-to-power mapping. The experiments evaluated how object weight affects the time required to reach stable lifting, the difference between lifting-estimation and releasing phases, and the suitability of progressive LCLE power regulation for energy-aware electromagnetic grasping.

B. Time-based evaluation criteria

Two time-based indicators were used: lifting estimation time and releasing time. The first measures the interval required for the Bayesian estimate to reach stable lifting while the LCLE power increases, and the second measures the time required for the estimated load to return near zero during object release. This distinguishes the weight-dependent lifting phase from the less load-dependent releasing phase.

C. Effect of object weight on lifting and return time

Fig. 4 compares the lifting estimation and releasing times for the evaluated object weights. The lifting estimation time increases with object weight, from approximately 10.9 s for 5 kg to 16.7 s for 9 kg, as heavier objects require a longer interval for the controller to estimate the load and reach sufficient LCLE power for stable lifting. In contrast, the releasing time remains within a narrower range, from approximately 3.6 s to 5.1 s, indicating that the Bayesian estimation strategy mainly affects the lifting phase, while the return phase is less dependent on object weight.

D. Representative estimation responses

Fig. 5 compares representative responses for the 5 kg and 9 kg objects, showing the corrected weight W_c, conservative prediction W_k^-, and Bayesian estimate $W_{\mathrm{Bayes},k}$ during lifting, holding, and return. The 5 kg case reaches a stable estimate earlier, while the 9 kg case requires a longer estimation interval. This agrees with the progressive power-regulation strategy, where the LCLE power command increases according to the corrected measurement and Bayesian estimate instead of applying maximum power at grasp onset.

The shaded regions in Fig. 5 indicate the lifting-estimation and releasing phases. The similar responses of W_c and $W_{\mathrm{Bayes},k}$ are expected under relatively stable measurement conditions. The Bayesian formulation does not necessarily produce a substantially different weight estimate; rather, it combines successive observations while explicitly representing estimation uncertainty. Unlike direct use of corrected sensor measurements, this formulation maintains a posterior distribution that supports the subsequent probabilistic weight-to-power mapping. However, the present results do not establish a quantitative advantage over direct sensor feedback or conventional filtering methods, which remains to be evaluated in future work.

E. Discussion of the control behavior

At the beginning of lifting, the measured load may be lower than the actual object weight because the object is still partially supported. As the load is transferred to the gripper, the conservative prediction prevents the predicted weight from falling below the current corrected measurement, while the Bayesian update combines this prediction with the prior estimate. Once the object is fully suspended, the estimated weight and power command converge to stable values, allowing the gripper to avoid maximum-power operation throughout the task. The main trade-off is an increase in lifting time, which is expected in a progressive estimation-based power-regulation strategy. Two-finger grippers are inherently sensitive to limited contact areas and off-center grasping, which may cause slipping or object rotation. These mechanical limitations were addressed in our previous work [1] and are outside the scope of the present study, which focuses on Bayesian weight estimation and progressive electromagnetic power regulation.

Nevertheless, controlled lifting is required to avoid insufficient holding force during load transfer.

VI. Discussion

The proposed framework provides a practical method for regulating electromagnetic grasping power from force-sensor measurements. The use of a piecewise calibration table compensates for load-dependent sensor bias, while Bayesian estimation reduces the influence of measurement uncertainty. In addition, the probabilistic conversion from estimated weight to power accounts for uncertainty around the estimated load. The main advantage of the method is that the object weight does not need to be known before grasping; it is estimated during lifting, and the electromagnetic power is increased progressively. Since both sensor correction and power assignment rely on empirical calibration, changes in object geometry, contact conditions, lifting speed, or battery voltage may affect estimation accuracy and required holding power. Therefore, the calibrated relationships are primarily valid under the tested operating conditions and may require recalibration for different configurations. Unlike the high-initial-power strategy reported in [1], which applies up to 529.2 W before reducing the input according to the object weight, the proposed method progressively increases the LCLE power based on the estimated load, avoiding an initially high-power command. The empirical calibration in Table I shows normalized commands of 7.5–11.7% for objects weighing 5–9 kg. Previous experiments also characterized winding temperatures under high-current operation [1]. Although the proposed strategy reduces the initial commanded power, its effect on total electrical energy consumption and winding temperature was not measured in this study.

VII. Conclusion

This paper presented a Bayesian weight-estimation method for power regulation in a liquid-cooled electromagnetic robotic gripper. The method combines load-cell-based weight measurement, empirical calibration, conservative load prediction, Bayesian estimation, and probabilistic weight-to-power conversion. Experiments with 5–9 kg objects showed that the method can estimate the load during lifting and adapt the normalized LCLE power command according to the estimated weight. The lifting-estimation interval was load-dependent, ranging from approximately 10.9 s to 16.7 s, while the releasing time remained within a narrower range of approximately 3.6–5.1 s. The proposed controller progressively increases the normalized LCLE power according to the estimated object weight, avoiding the high initial power command used in the previous control strategy.

Future work will quantitatively compare the proposed method with direct sensor feedback, conventional filtering, and high-initial-power control in terms of estimation accuracy, energy consumption, and grasp stability. Additional experiments will also evaluate the effect of object shape, handle diameter, lifting speed, and temperature variation on the estimated weight and required power.

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References

[1] M. R. Calvo, T. Nishimura, and T. Watanabe, "Flexible liquid-cooled electromagnetic gripper for grasping heavy objects," *IEEE Access*, vol. 13, pp. 133370–133380, 2025.

[2] M. Rodriguez, T. Nishimura, and T. Watanabe, "LCLE: A liquid-cooled laminar electromagnet for a soft robotic gripper and heavy object grasping," in *Proc. IEEE 5th Int. Conf. BioInspired Processing (BIP)*, Nov. 2023, pp. 1–7.

[3] S. Marullo, G. Salvietti, and D. Prattichizzo, "On the use of magnets to robustify the motion control of soft hands," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 11743–11750, Oct. 2022.

[4] S. Marullo, S. Bartoccini, G. Salvietti, M. Z. Iqbal, and D. Prattichizzo, "The Mag-Gripper: A soft-rigid gripper augmented with an electromagnet to precisely handle clothes," *IEEE Robotics and Automation Letters*, vol. 5, no. 4, pp. 6591–6598, Oct. 2020.

[5] M. Dragusanu, S. Marullo, M. Malvezzi, G. M. Achilli, M. C. Valigi, D. Prattichizzo, and G. Salvietti, "The dress gripper: A collaborative gripper with electromagnetic fingertips for dressing assistance," *IEEE Robotics and Automation Letters*, vol. 7, no. 3, pp. 7479–7486, Jul. 2022.

[6] Y. Tsugami and T. Nishida, "Simple structured gripper using electromagnet and permanent magnet," in *Proc. Int. Conf. ICT Robotics*, vol. 2017, Nov. 2017, pp. 25–26.

[7] Y. J. Shin, K.-H. Rew, K. Kim, and S. Kim, "Development of anthropomorphic robot hand with dual-mode twisting actuation and electromagnetic joint locking mechanism," in *Proc. IEEE Int. Conf. Robotics and Automation (ICRA)*, May 2013, pp. 2759–2764.

[8] Y. J. Shin, H. J. Lee, K.-H. Rew, K.-S. Kim, and S. Kim, "Development of electromagnet brake robot finger for highly dexterous motion through a single motor," in *Proc. 12th Int. Conf. Control, Automation and Systems (ICCAS)*, Oct. 2012, pp. 1423–1426.

[9] S. Cheng, W. Zhang, and W. Li, "A novel underactuated robot finger with a limit of electromagnet braking for parallel pinching, coupling and self-adaptive grasps," in *Proc. 5th Int. Conf. Advanced Robotics and Mechatronics (ICARM)*, Dec. 2020, pp. 84–89.

[10] R. Liang, G. Xu, M. Li, S. Zhang, A. Luo, and T. Tao, "A novel variable stiffness compliant finger exoskeleton for rehabilitation based on electromagnet control," in *Proc. 40th Annual Int. Conf. IEEE Engineering in Medicine and Biology Society (EMBC)*, Jul. 2018, pp. 3926–3929.

[11] T. Kamf and J. Abrahamsson, "Self-sensing electromagnets for robotic tooling systems: Combining sensor and actuator," *Machines*, vol. 4, no. 3, Art. no. 16, Aug. 2016.

[12] T. Le Signor, N. Dupré, and G. F. Close, "A gradiometric magnetic force sensor immune to stray magnetic fields for robotic hands and grippers," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 3070–3076, Apr. 2022.

[13] A. C. Holgado, T. P. Tomo, S. Somlor, and S. Sugano, "A multimodal, adjustable sensitivity, digital 3-axis skin sensor module," *Sensors*, vol. 20, no. 11, Art. no. 3128, Jun. 2020.

[14] X. Li, C. Lu, Z. Song, W. Ding, and X. P. Zhang, "Planar magnetic actuation for soft and rigid robots using a scalable electromagnet array," *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 9264–9270, Oct. 2022.

[15] D. Liu, X. Liu, P. Li, X. Tang, M. Kojima, Q. Huang, and T. Arai, "All-purpose magnetic micromanipulation system with two modes: Chopstick-like two-finger microhand and hydrodynamic tweezer," *IEEE/ASME Transactions on Mechatronics*, doi: 10.1109/TMECH.2021.3090661.

[16] T. Nahmura and I. Ando, "Design of micro-hand with two magnetic levitated fingers," in *Proc. Int. Symp. Micromechatronics and Human Science*, 1997, pp. 137–142.

[17] S. Thrun, W. Burgard, and D. Fox, *Probabilistic Robotics*. Cambridge, MA, USA: MIT Press, 2005.